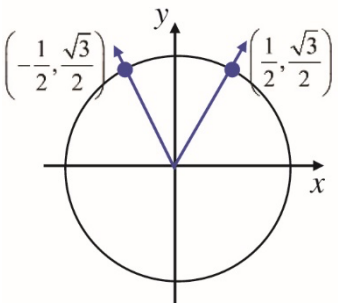
Solve Trigonometric Equations in Linear Form

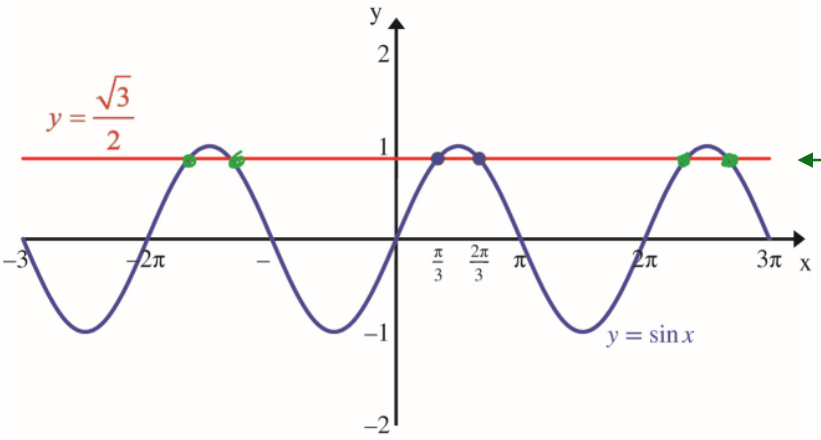
Example: Solve. $\sin x = \frac{\sqrt{3}}{2}$

From the unit circle, we can see that there are 2 values of x between 0 and 2π that will solve the equation: $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$.



However, there are actually infinitely many solutions to the equation:

$x = \frac{\pi}{3} + 2n\pi$ and $x = \frac{2\pi}{3} + 2n\pi$ (where n is any integer).



all intersections are solutions

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1. Solve the equation on the interval $[0, 2\pi)$. $3\csc x = 4 + \csc x$

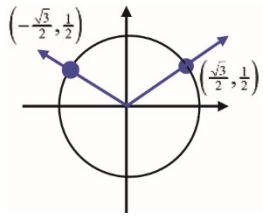
$3\csc x = 4 + \csc x$

$2\csc x = 4$

$\csc x = \frac{4}{2}$

$\sin x = \frac{2}{4} \rightarrow \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$



2. Solve the equation over the set of real numbers. $\sqrt{2}\cos x + 5 = 4$

$\sqrt{2}\cos x + 5 = 4$

$\sqrt{2}\cos x = -1$

$\cos x = \frac{-1}{\sqrt{2}}$

$\cos x = -\frac{\sqrt{2}}{2}$

$x = \frac{3\pi}{4} + 2n\pi$ & $\frac{5\pi}{4} + 2n\pi$

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Solve Trigonometric Equations Involving Multiple or Compound Angles

Solve the equation on the interval $[0, 2\pi)$.

3. $8\cot 2x - 5 = 3$

Isolate $\cot 2x$.

Give the general solution from the unit circle.

To find solutions on the interval $[0, 2\pi)$, substitute $n = \ldots, -1, 0, 1, 2, \ldots$

$8\cot 2x - 5 = 3 \quad \cot = \frac{\cos}{\sin}$

$8\cot 2x = 8$

$\cot 2x = 1$

$2x = \frac{\pi}{4}, \frac{5\pi}{4}$

$x = \frac{\pi}{8} + n\pi \quad \frac{5\pi}{8} + n\pi$

$x = \frac{\pi}{8}, \frac{9\pi}{8}, \frac{5\pi}{8}, \frac{13\pi}{8}$

4. $\cos\frac{x}{2} = \frac{\sqrt{3}}{2}$

$\cos\frac{x}{2} = \frac{\sqrt{3}}{2}$

$\frac{x}{2} = \frac{\pi}{6}, \frac{11\pi}{6}$

$\frac{x}{2} = \frac{\pi}{6} + 2n\pi$

$x = \frac{\pi}{3} + 4n\pi$

$\frac{x}{2} = \frac{11\pi}{6} + 2n\pi$

$x = \frac{22\pi}{6} + 4n\pi$

$x = \frac{\pi}{3}$

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Solve Higher-Degree Trigonometric Equations

Solve the equation on the interval $[0, 2\pi)$.

5. $2\sin^2 x + 5\sin x + 3 = 0$

$2u^2+5u+3=0$

$(2u^2+2u)+(3u+3)$

$2u(u+1)3(u+1)$

$(2u+3)(u+1)$

$2\sin x+3=0$

$2\sin x=-3$

$\sin x=\frac{-3}{2}$

no solution

$\sin x+1=0$

$\sin x=-1$

$x=\frac{3\pi}{2}$

$x=\frac{3\pi}{2}+2n\pi$

$x=\frac{3\pi}{2}$

Use Identities to Solve Trigonometric Equations

Solve on the interval $[0, 2\pi)$.

6. sin 2x = cos x

$\sin 2x=\cos x$

$2\sin x\cdot\cos x=\cos x$

$2\sin x\cdot\cos x-\cos x=0$

$\cos x(2\sin x-1)$

$\cos x=0$

$x=\frac{\pi}{2},\frac{3\pi}{2}$

$2\sin x-1=0$

$2\sin x=1$

$\sin x=\frac{1}{2}$

$x=\frac{\pi}{6},\frac{5\pi}{6}$

7. sin x = cos x

$\sin x=\cos x$

$\frac{\sin x}{\cos x}=1$

$\tan x=1$

$x=\frac{\pi}{4},\frac{5\pi}{4}$

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Use Inverse Functions to Express Solutions to Trigonometric Equations

8. Solve on the interval $[0, 2\pi)$. $4\cos^2 x - 7\cos x + 3 = 0$

$4\cos^2 x - 7\cos x + 3 = 0$

$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$\frac{7 \pm \sqrt{49 - 4(4)(3)}}{2(4)}$

$\frac{7 \pm \sqrt{1}}{8}$

$\cos x = 1$

$x = 0$

$\cos x = \frac{3}{4}$

$x = \cos^{-1}\left(\frac{3}{4}\right)$

$x = .723$

$+$ $-$

$\frac{7+1}{8}$ $\frac{7-1}{8}$

$\frac{8}{8}$ $\frac{6}{8}$

1 $\frac{3}{4}$

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